

Computer Vision Lecture 5

Elements of Deep Learning (Part 2)

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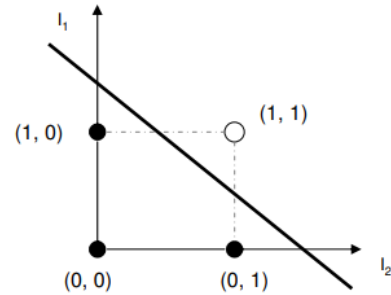


Recap

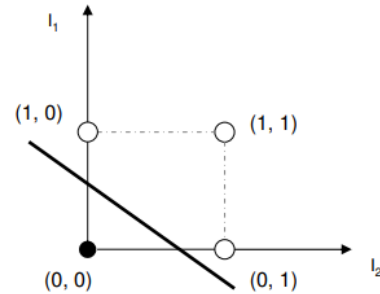
- Overcoming the limitations of linear models is important
- Neural networks provide an effective way of generalizing linear models
- Gradient based learning can be used for parameter estimation

Learning XOR

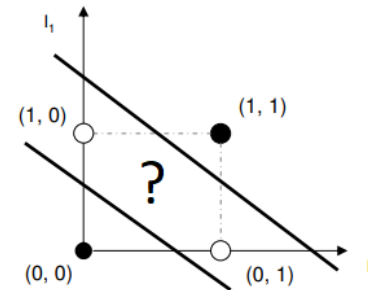
AND		
I_1	I_2	out
0	0	0
0	1	0
1	0	0
1	1	1



OR		
I_1	I_2	out
0	0	0
0	1	1
1	0	1
1	1	1



XOR		
I_1	I_2	out
0	0	0
0	1	1
1	0	1
1	1	0

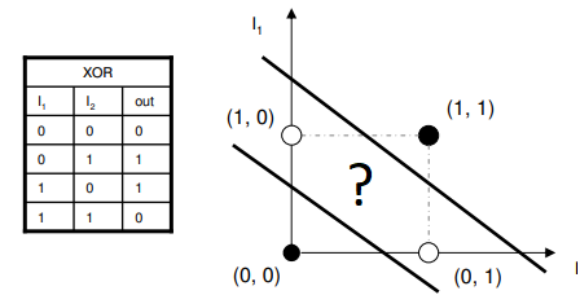


Can we approximate or learn the XOR function with a linear model?

Learning XOR

Evaluated on the entire training set, the MSE loss function is:

$$J(\theta) = \frac{1}{4} \sum_x (f^*(x) - f(x; \theta))^2$$



Training Set

Our model is of the form:

$$f(x; w, b) = x^T w + b$$



$$w = 0 \quad \text{and} \quad b = \frac{1}{2}$$

Learning XOR

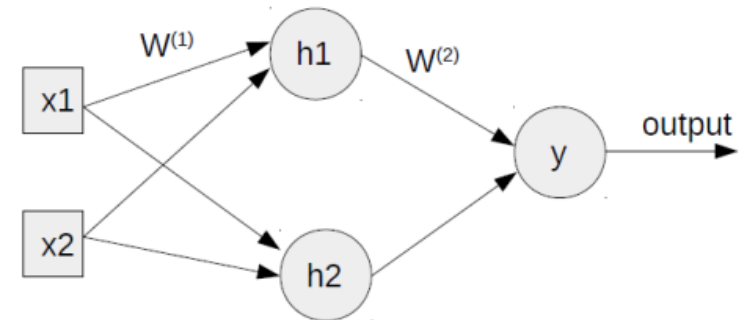
Hidden units:

$$h_i = f^{(1)}(x; W_i, c_i)$$

Output units:

$$y = f^{(2)}(h; w, b)$$

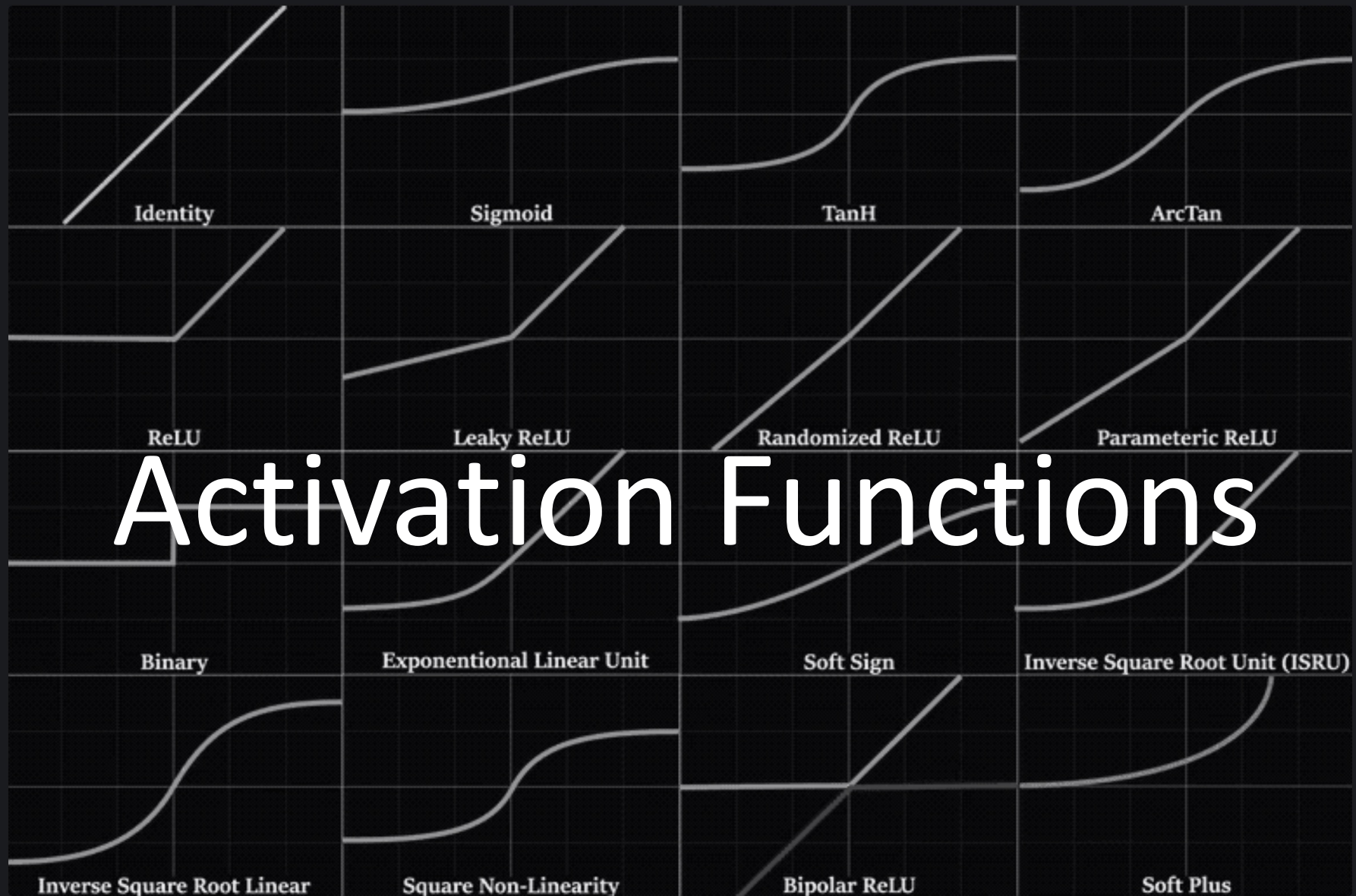
We now have to find a nonlinear function $f^{(1)}$ That allows us to approximate the XOR function.



Feedforward network with on hidden layer containing two units

[Link](#) to web page and programming example.

Activation Functions



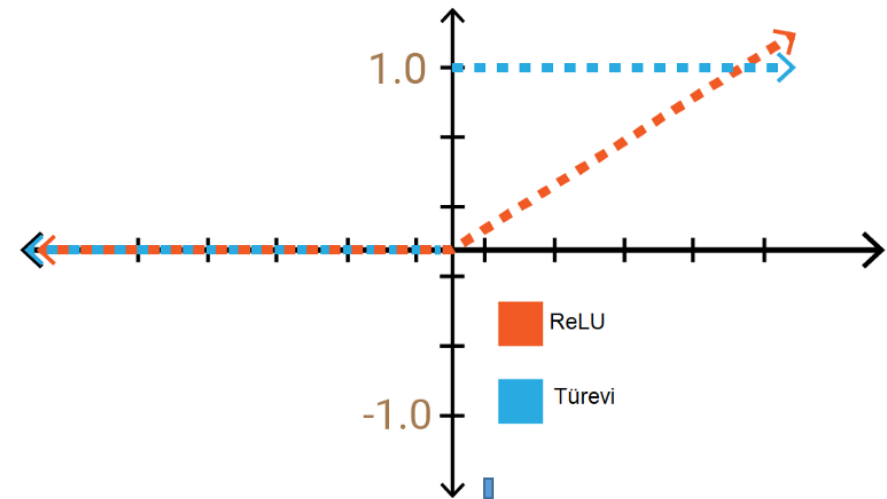
ReLU

Rectified linear unit (ReLU):

$$g(z) = \max \{0, z\}$$

Our complete XOR network can now be written as:

$$f(x; W, c, w, b) = w^T \max\{0, W^T x + c\} + b$$



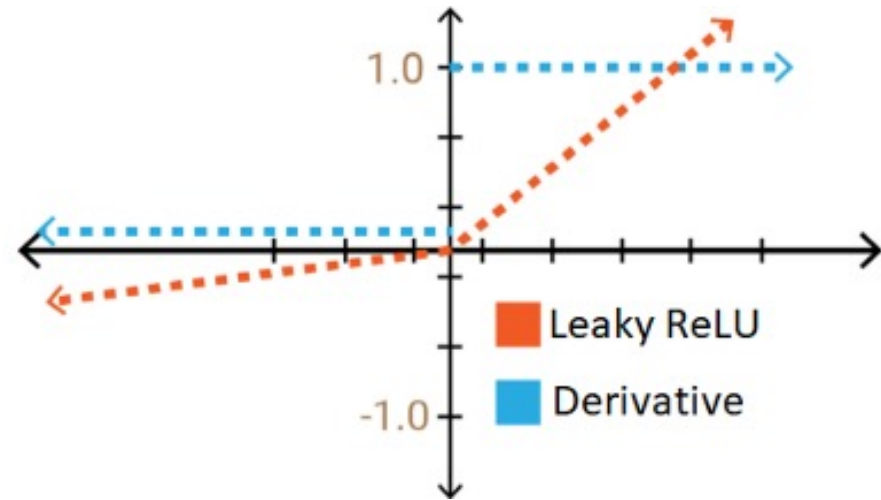
$$W = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad c = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$
$$w = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad b = 0$$

ReLU

- Compared to a sigmoid function the ReLU can be computed faster
- But, we also don't learn anything when $g(z)$ is zero

Leaky-ReLU

$$f(x) = \begin{cases} x & \text{if } x > 0, \\ 0.01x & \text{otherwise.} \end{cases}$$

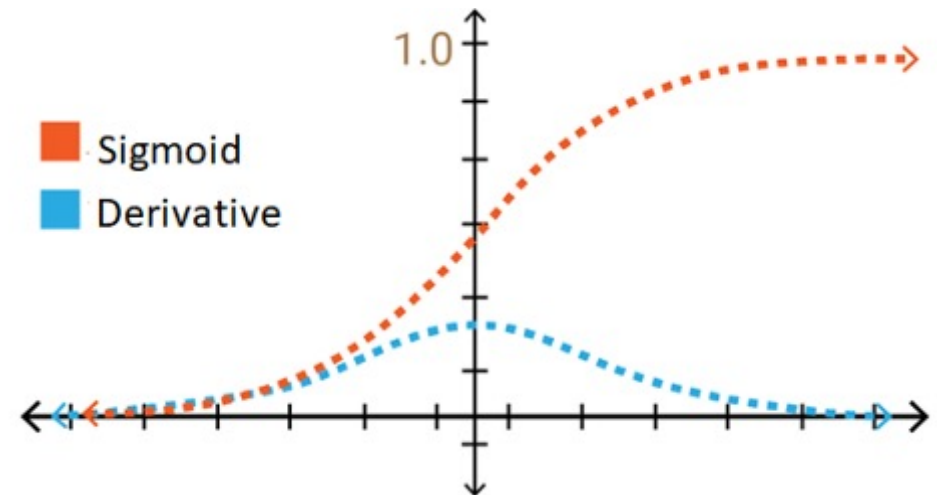


There are a number of other variants of the Leaky-ReLU activation function.

Sigmoid function

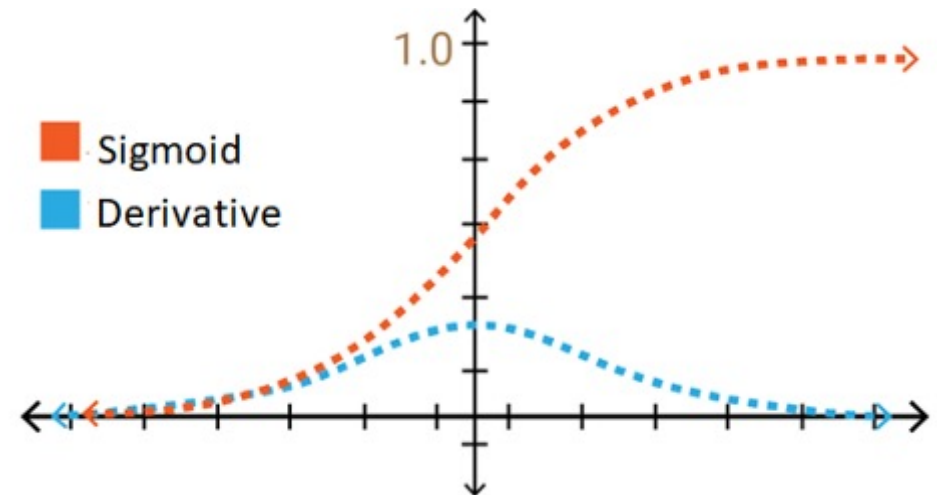
Definition:

$$f(x) = \sigma(x) = \frac{1}{1 + e^{-x}}$$



Sigmoid function

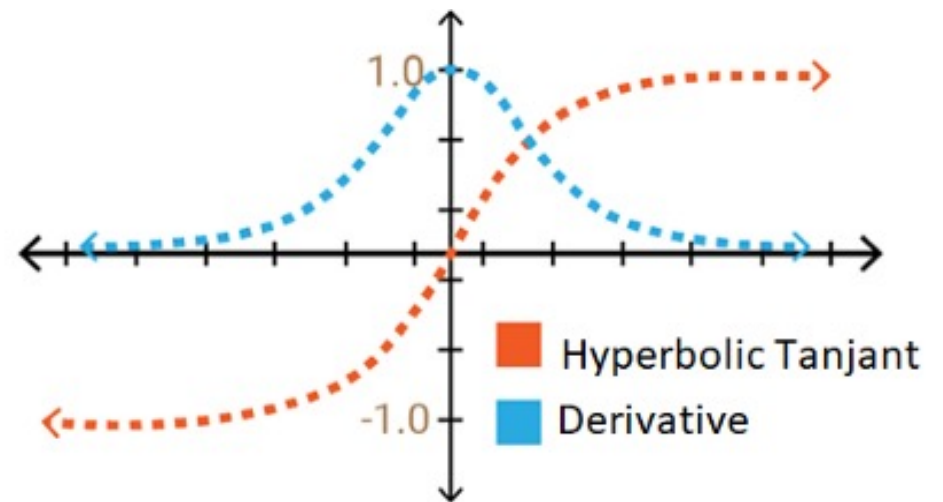
- Positive: the activation function does not vanish
- Negative: the vanishing gradient means can imply that learning gets stuck in local minima



Hyperbolic tangent

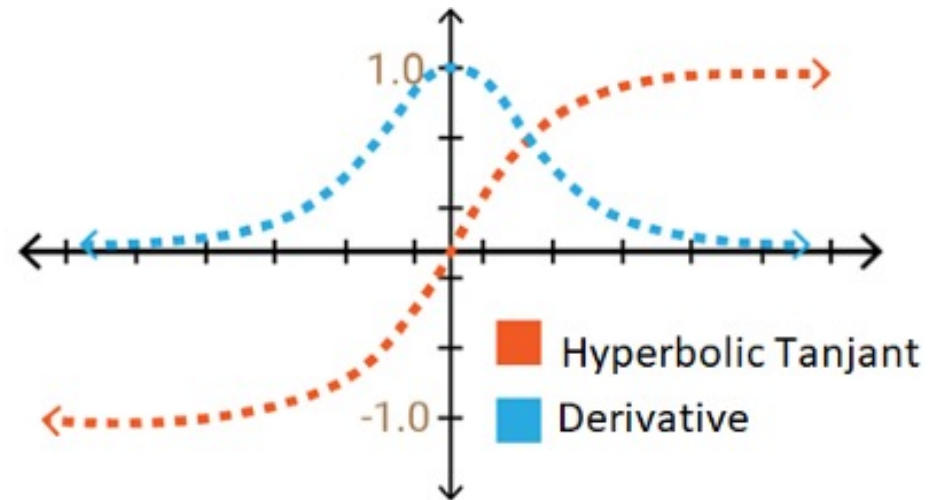
Definition:

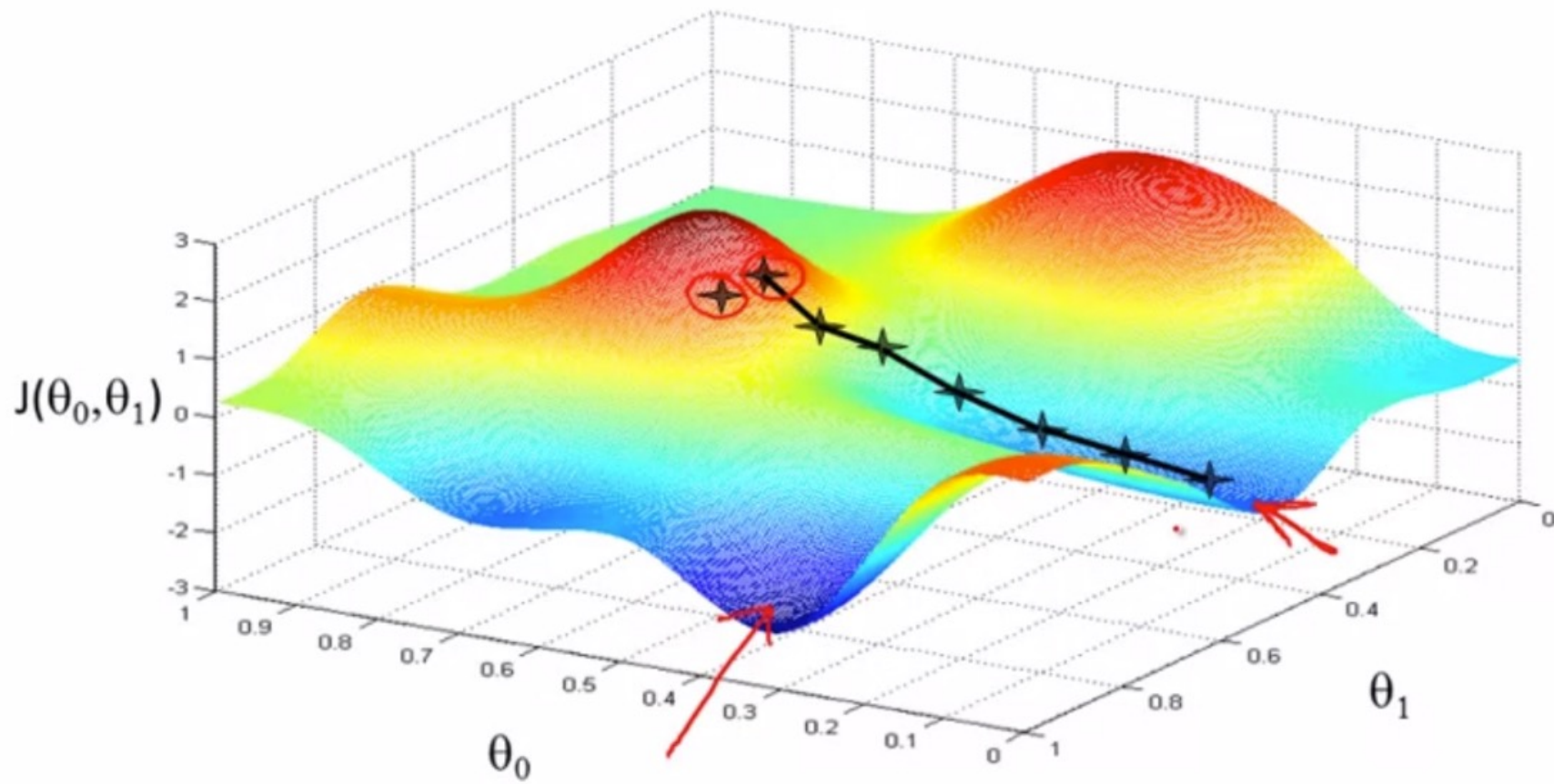
$$f(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Hyperbolic tangent

- Compared to the sigmoid it has steeper gradient
- But we still have the problem of vanishing gradients





Cost Functions

Cost functions

- We can use the same cost functions we have used for parametric models for neural networks
- Our model defines a distribution

$$p(y|x; \theta)$$

and we can identify the model parameters by optimizing the cross entropy between the training data and the model's predictions

Cost functions

- Most modern neural networks are trained using maximum likelihood:

$$J(\theta) = -E_{x,y} \log p_{model}(y|x)$$

- The gradient of the cost function must be large enough to serve as a good guide for the learning algorithm

Output Units

- The role of the output layer is to provide an additional transformation from the features to compete the prediction task
- The choice of the output units is tightly coupled with the choice of the cost function

Output units: linear units

- Simple output unit that is based on an affine transform and has no nonlinearity

$$p(y|x) = N(y, \hat{y}, I) \qquad \hat{y} = W^T h + b$$

- Maximizing the log-likelihood is then equivalent to minimizing the mean squared error

Output units: sigmoid output unit

- The sigmoid output unit is defined by

$$\hat{y} = \sigma(w^T h + b)$$

- The unit uses a sigmoid activation function to convert z into a probability
- It is particularly useful for predicting binary variables



Architectures

Architectures

- The universal approximation theorem tells us that a feedforward network with a single hidden layer can approximate a broad class of functions provided it has enough hidden units.
- A network with a single layer might be infeasibly large and may fail to learn and generalize properly
- Decomposing the network into layers amounts to decomposing the function we are trying to approximate

Architectures

- Empirical results show that deeper networks generalize better
- However, it is difficult to provide general guidance – particular strategies depend on the application context

Optimization algorithms

- We reviewed back-propagation for traditional neural networks
- The goal of machine learning is to reduce the expected generalized error (also known as risk)
- In practice we minimize the expected loss on the training set, known as the empirical risk

$$\frac{1}{m} \sum_i L(f(x^i; \theta), y^i)$$

Optimization algorithms

- Realistically we cannot compute the gradients for the entire training set
- We estimate the gradient from a number of samples assuming that there is a redundancy in the training set.
- Optimization methods estimate the gradients using random subsets of the data are called minibatch stochastic models or simply stochastic models.

Optimization methods

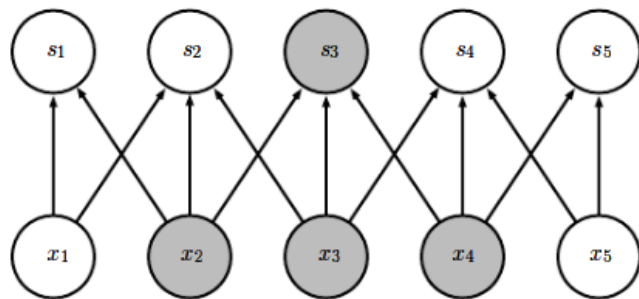
- A range of stochastic optimization algorithms exists
 - Stochastic Gradient Decent (SGD)
 - Algorithms with adaptive learning rates: ADAM
- There is not theoretical results which of these is the best. Users typically pick the algorithm they are most familiar with.

Convolutional networks

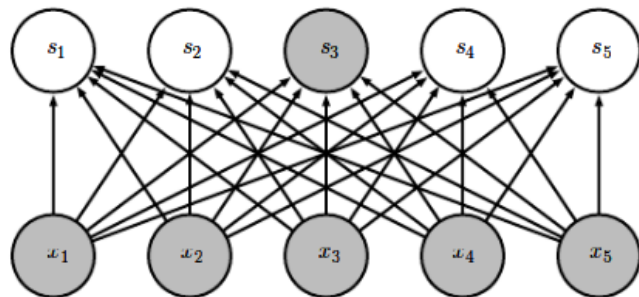
- Convolutional networks are networks that use convolution in place of general matrix multiplication in at least one of the layers
- They are particularly relevant to image analysis and signal processing problem
- Here the convolution operation you have already learnt is being used. However, the parameters of the filter or kernel are being learnt directly from the training data.

Convolutional network

Convolutional layers implement sparse interactions



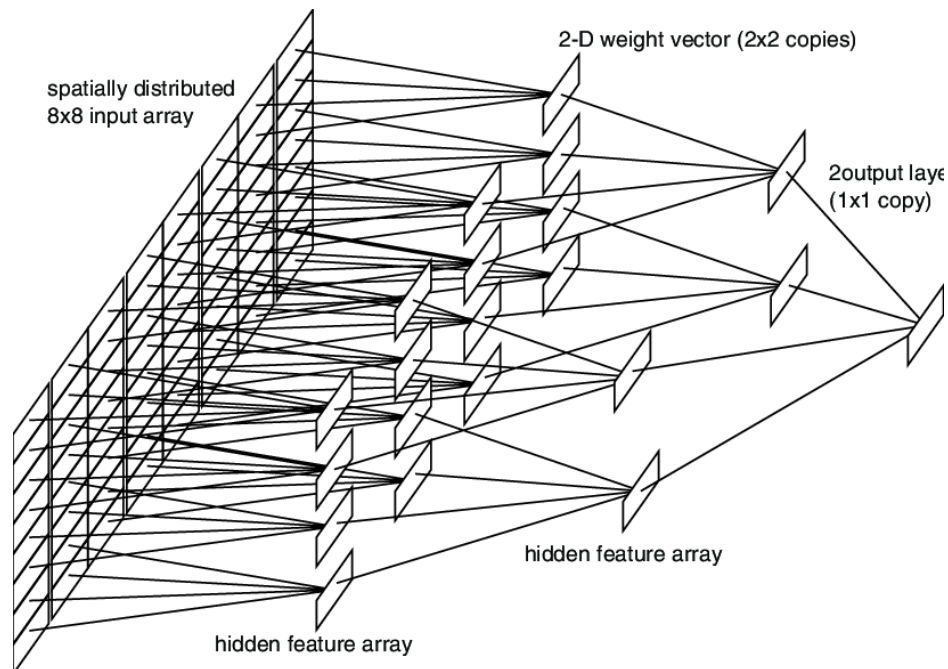
Convolution



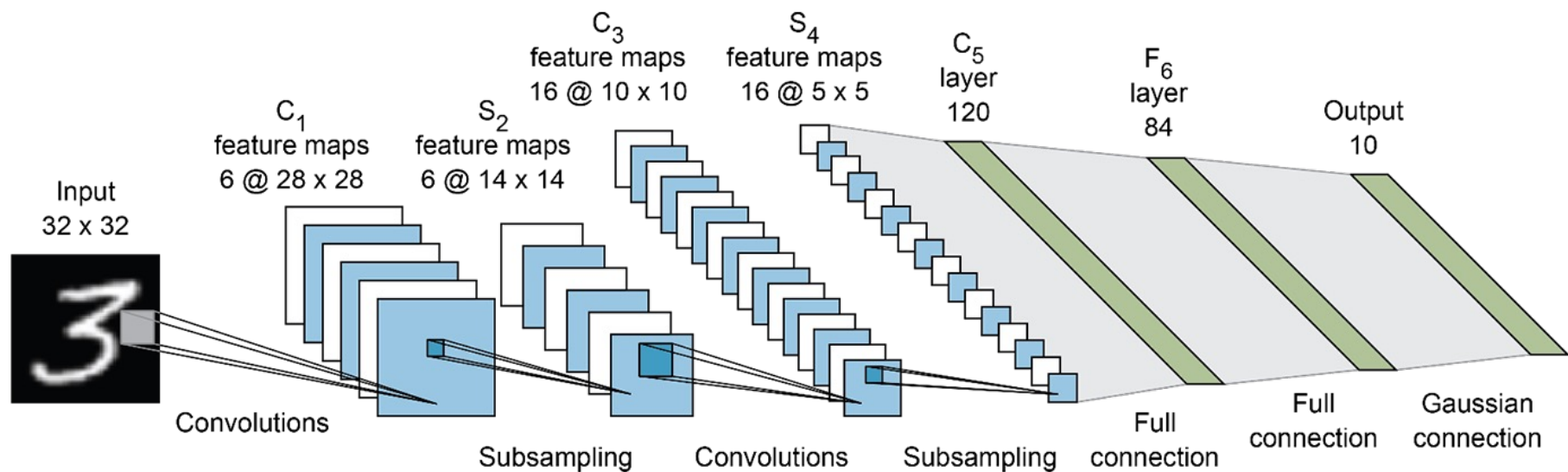
Matrix multiplication

Convolutional networks

Through the hierarchical architecture the network effectively interact with a large portion of the input



LeNet: A layered model for the classification of digits

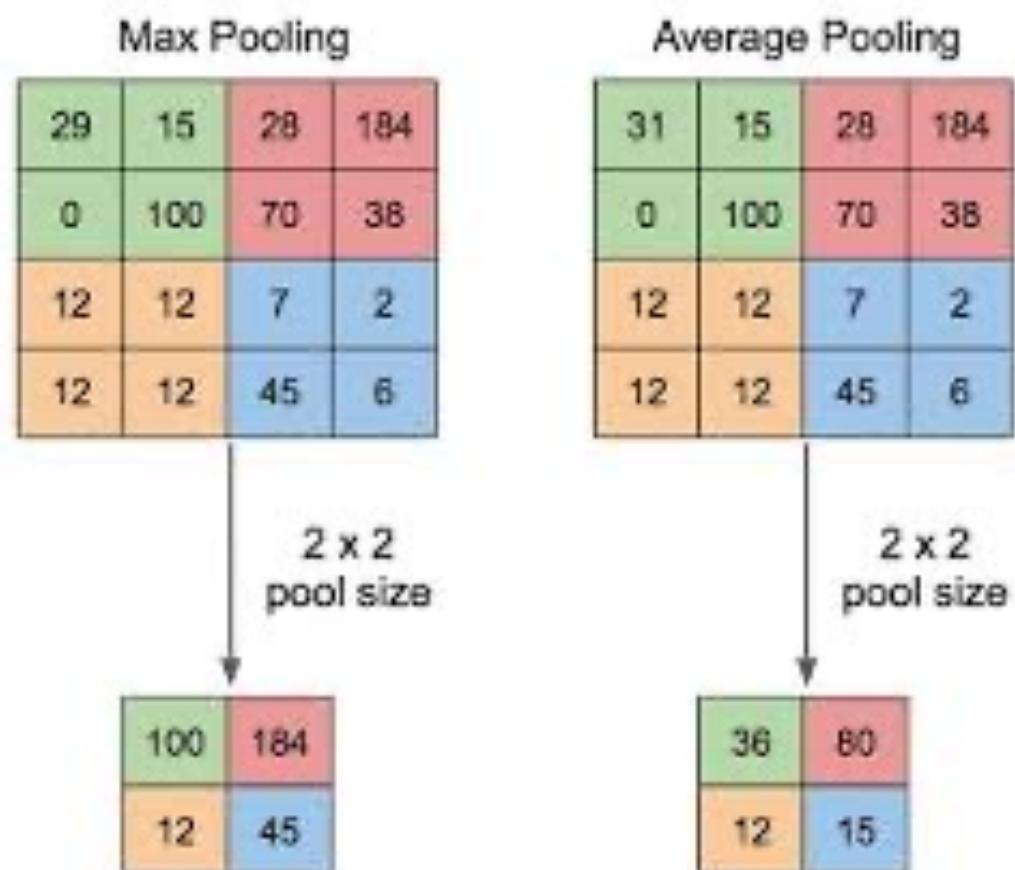


LeCun, Y., Boser, B., Denker, J.S., Henderson, D., Howard, R.E., Hubbard, W. and Jackel, L.D., 1989. Backpropagation applied to handwritten zip code recognition. *Neural computation*, 1(4), pp.541-551.

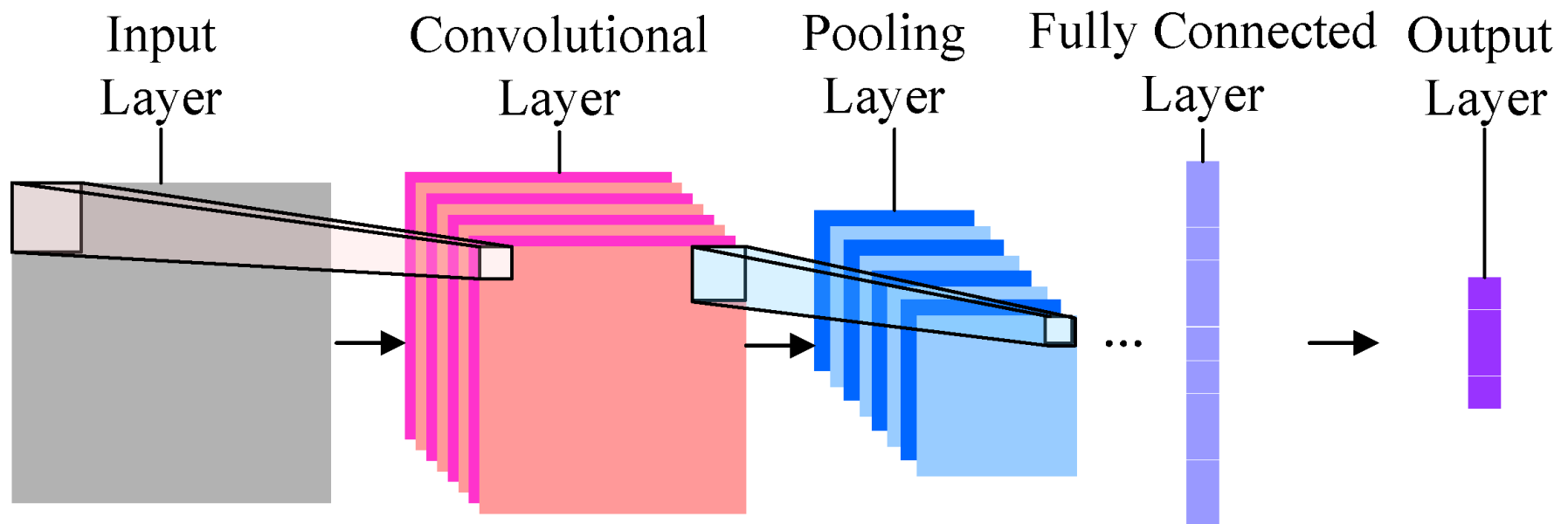
Pooling operation

- A pooling function replaces the output of the network at a certain location with a summary statistic of the nearby outputs
- Pooling helps to make the representation approximately invariant to small translations of the input

Pooling



A typical convolution layer



ImageNet – Large Scale Visual Recognition Challenge

The challenge evaluates algorithms for object localization and detection in videos. It includes the following tasks:

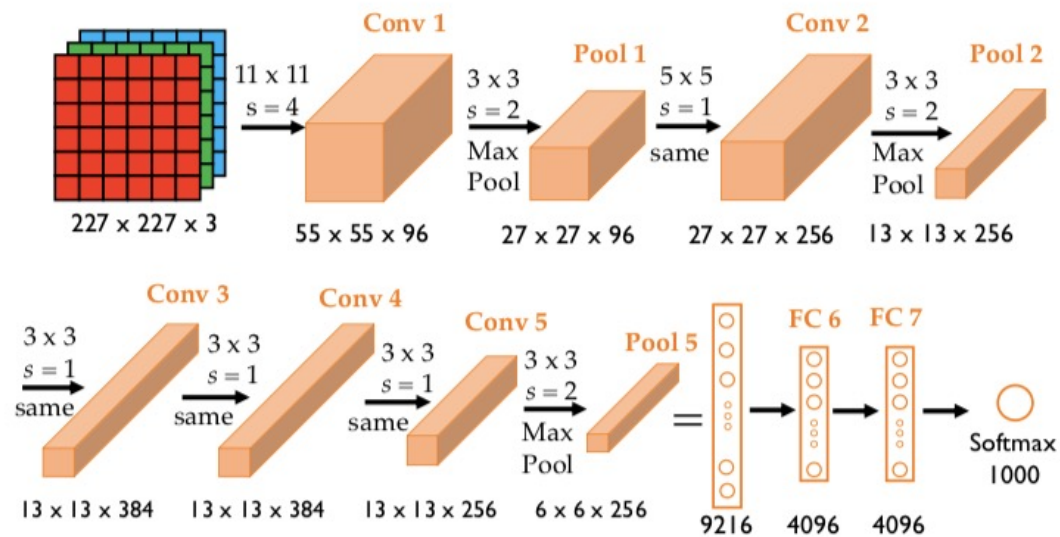
- Object localization for 1000 categories
- Object detection for 200 fully labelled categories
- Object detection from video for 30 fully labelled categories

The data consists millions for hand annotated images taken from flickr and other search engines



AlexNet

- Winner of the 2012 ImageNet Large Scale Recognition Challenge



Krizhevsky, A., Sutskever, I. and Hinton, G.E., 2012. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems* (pp. 1097-1105).

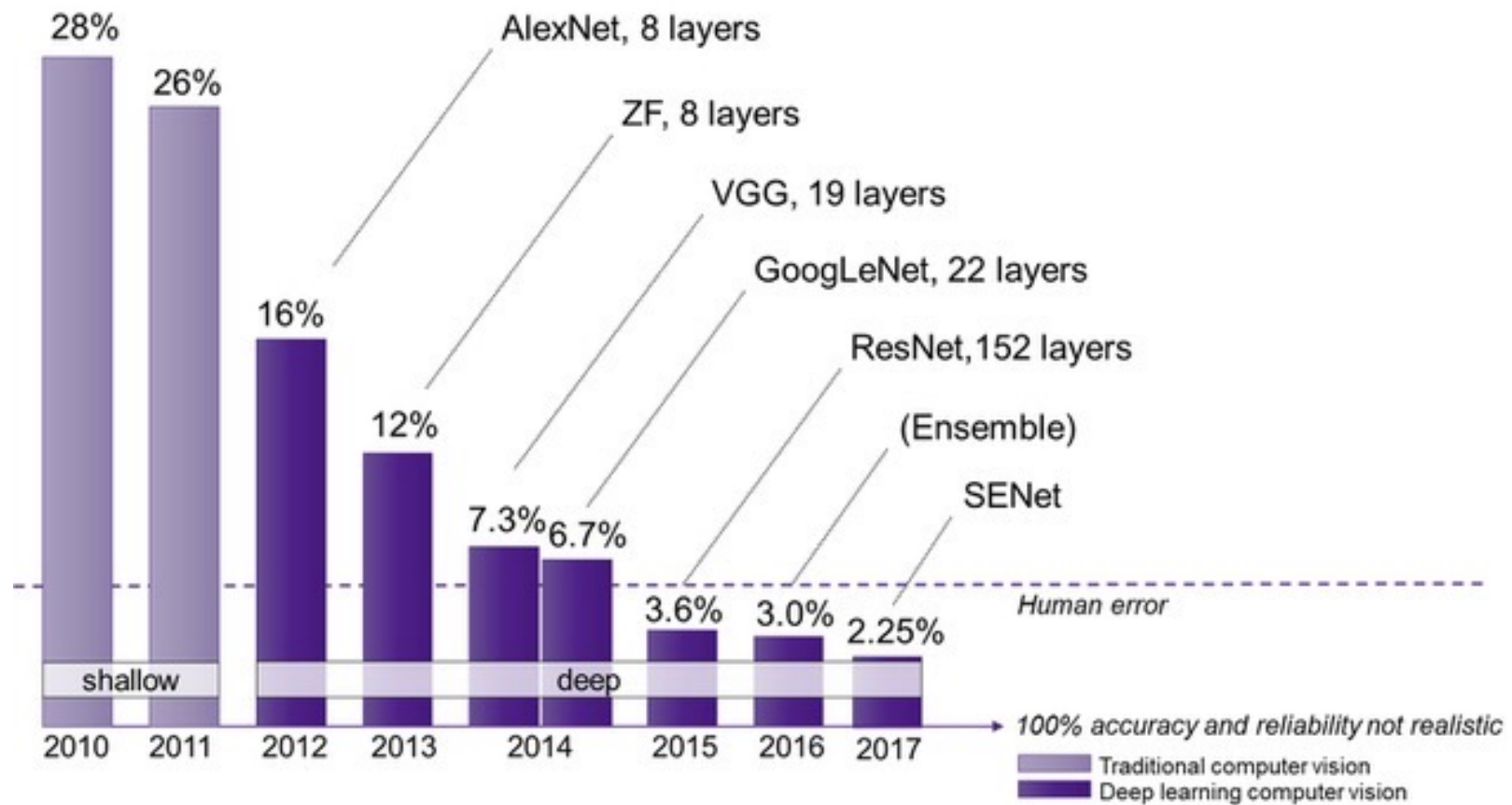
VGG Net Architectures

1st runner up in the ImageNet Challenge in 2014

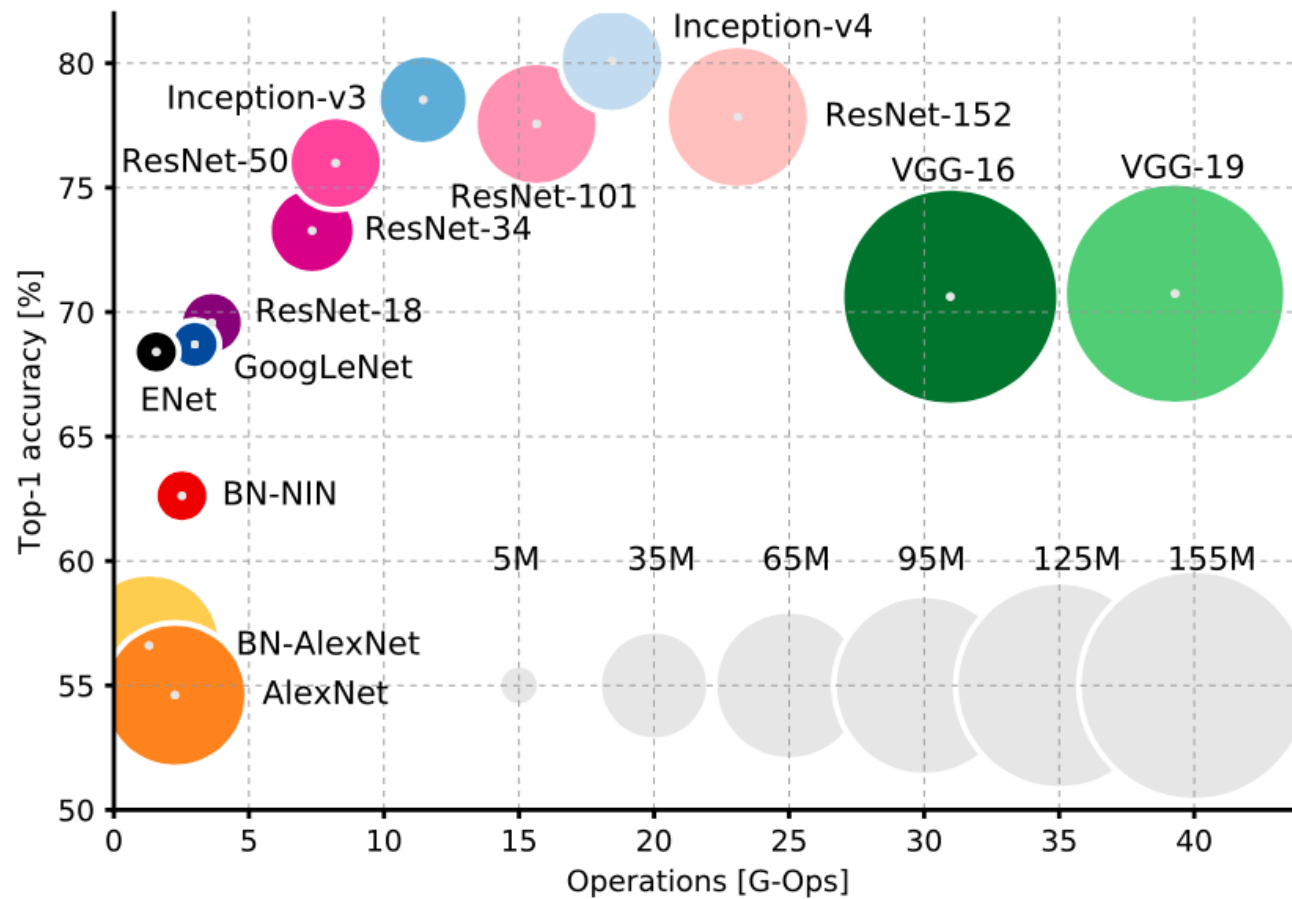
		Number of Parameters (millions)	Top-5 Error Rate (%)
		133	10.4
		133	10.5
		133	9.9
		134	9.4
		138	8.8
		144	9.0

Simonyan, K. and Zisserman, A., 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*.

Summary



Size vs complexity



Visualising features

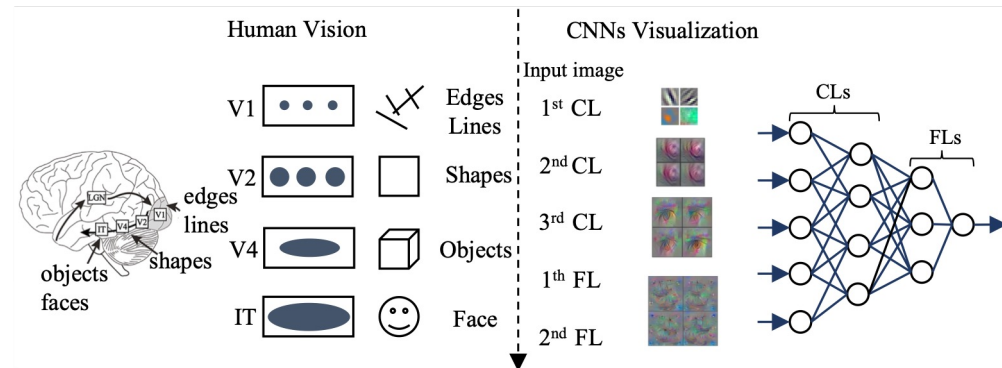
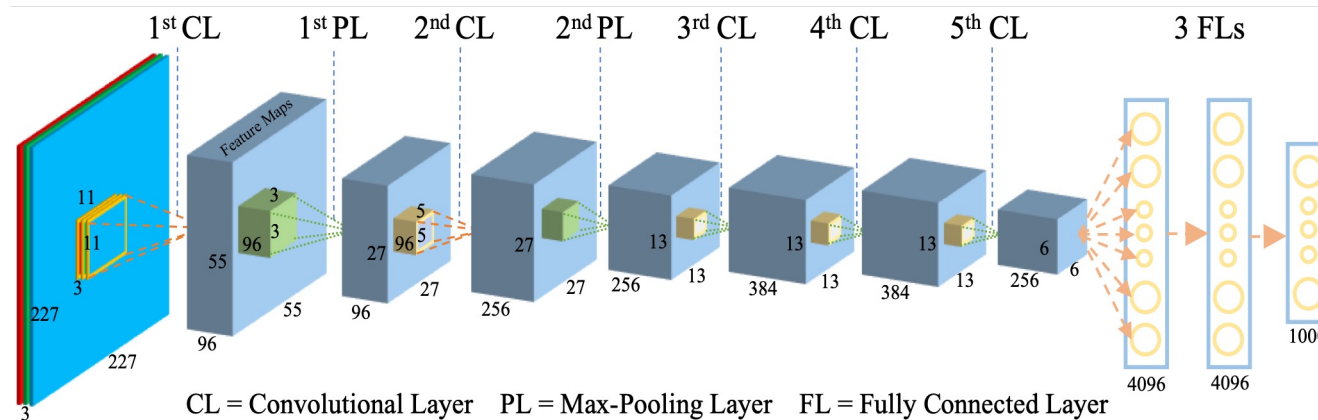
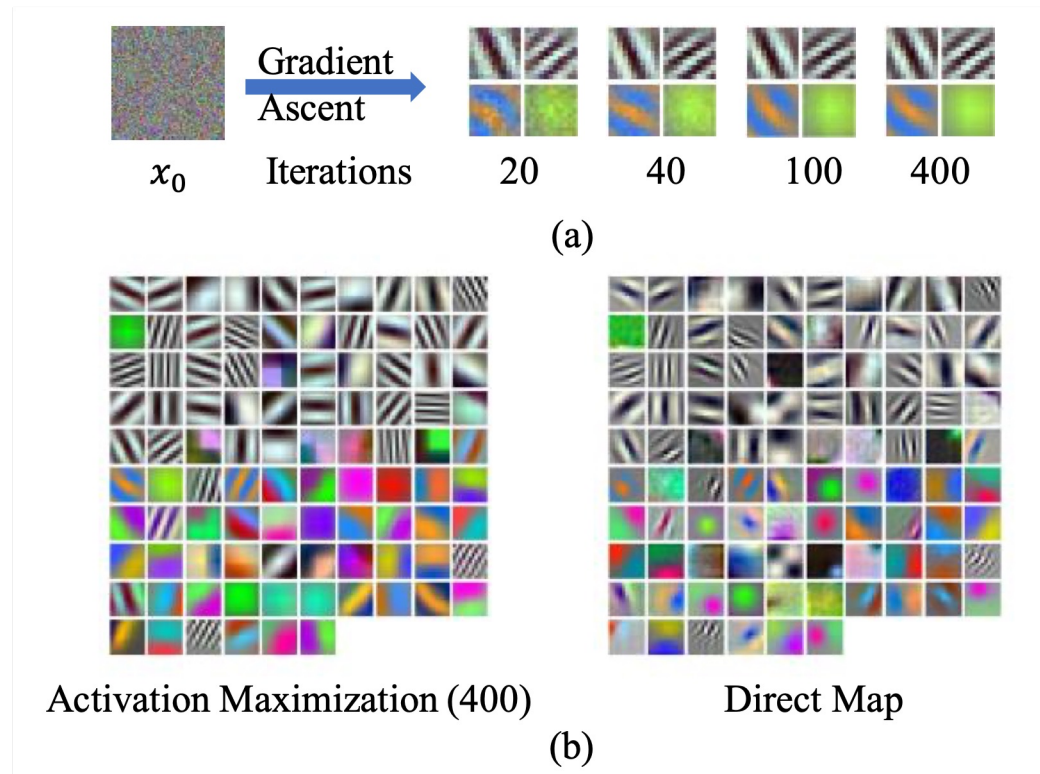


FIGURE 3. Human vision and CNNs visualization

Visualising Layer 1



Visualising layers

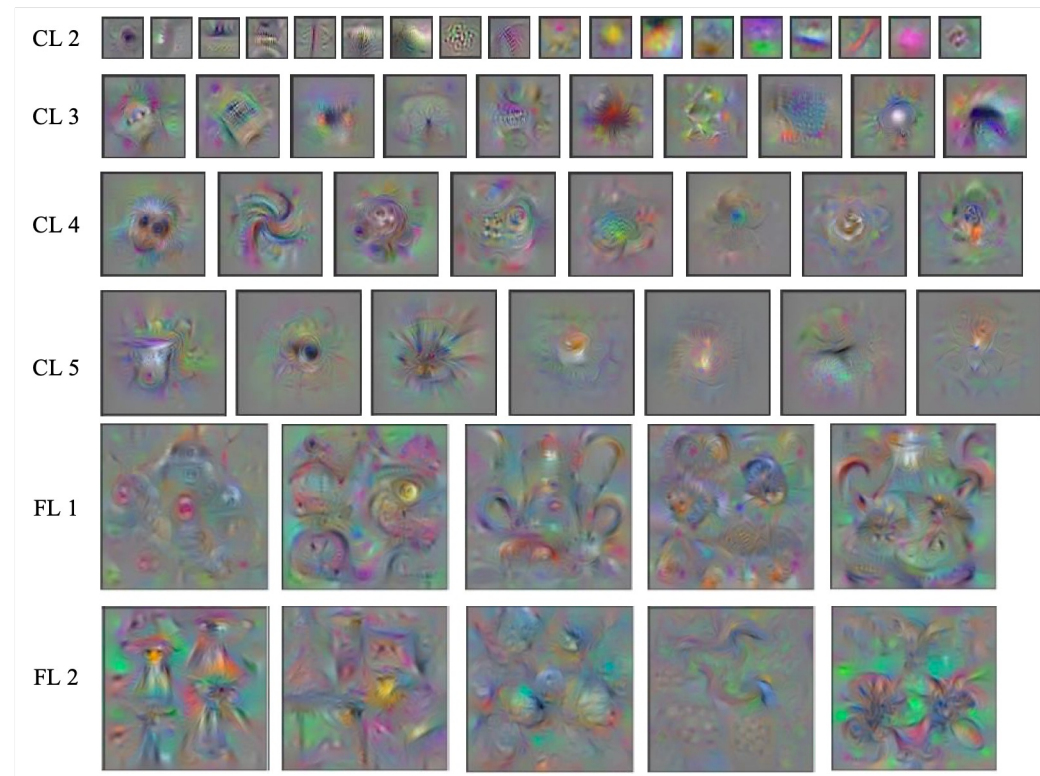
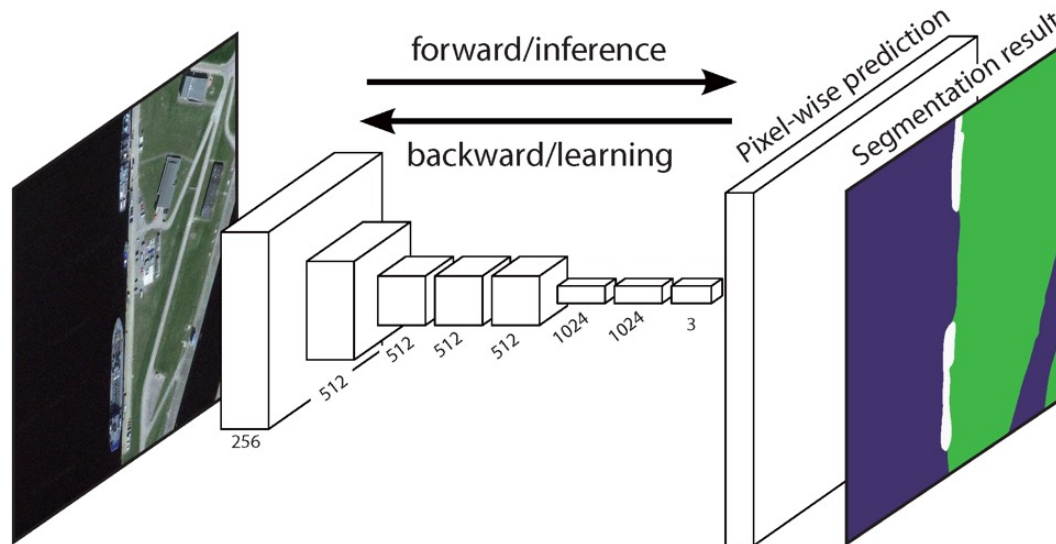


FIGURE 5. Hidden layers of *CaffeNet* visualization by *Activation Maximization*. Adapted from “Understanding Neural Networks Through Deep Visualization,” by J. Yosinski, 2015.

Semantic segmentation – fully convolutional networks

The original convolutional network learns a mapping from pixels to pixels via an upsampling step



Long, J., Shelhamer, E. and Darrell, T., 2015. Fully convolutional networks for semantic segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 3431-3440).

Fully convolutional networks

